MATH 6610: Advanced Numerical Analysis

Homework 3

Due: Friday, November 6, 2026

Instructions. Show all work. Proofs should be complete but may cite results proved in lecture.

Problem 1. Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$.

(a) Compute $\|A\|_1$, $\|A\|_\infty$, and $\|A\|_F$.

(b) Prove the general inequality $\|A\|_2 \le \sqrt{\|A\|_1 \, \|A\|_\infty}$ for any square A. (Hint: $\|A\|_2^2 = \lambda_{\max}(A^H A)$, and the spectral radius of a matrix is bounded by any induced norm of it.)

(c) Use (a) and (b) to give an upper bound for $\|A\|_2$, and compare with the true value $\|A\|_2 = \sqrt{15 + \sqrt{125}} \approx 5.116$.

Problem 2. Let $\delta > 0$ and

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1+\delta \end{bmatrix}.$$

(a) Compute A^{-1} and the condition number $\kappa_\infty(A)$ exactly as a function of δ, and evaluate it for $\delta = 10^{-6}$.

(b) Solve $Ax = b$ by hand for $b = (2,\, 2+\delta)^T$ and for $\tilde{b} = (2, 2)^T$.

(c) Compute the relative change of the data, $\left\|b - \tilde{b}\right\|_\infty / \|b\|_\infty$, and of the solutions, and verify that their ratio is bounded by (and here comparable to) $\kappa_\infty(A)$.

Problem 3. (Backward stability of the inner product.) Let $x, y \in \mathbb{R}^n$ and consider computing $s = x^T y$ by the recursive summation $s := \mathrm{fl}(\cdots \mathrm{fl}(\mathrm{fl}(x_1 y_1) + \mathrm{fl}(x_2 y_2)) + \cdots + \mathrm{fl}(x_n y_n))$. Using the fundamental axiom (each operation exact up to a factor $1+\epsilon$, $|\epsilon| \le \frac{1}{2}\varepsilon_{\mathrm{mach}}$), show that the computed value satisfies

$$\mathrm{fl}(x^T y) = \sum_{i=1}^{n} x_i y_i (1+\theta_i) \qquad \text{with} \qquad |\theta_i| \le n\,\varepsilon_{\mathrm{mach}} + O(\varepsilon_{\mathrm{mach}}^2),$$

and explain why this identity exhibits the algorithm as backward stable (hint: absorb $1+\theta_i$ into x_i).

Problem 4. Let $p \in \prod_3$ interpolate $f(x) = 2^x$ at the nodes $x = 0, 1, 2, 3$ (values $1, 2, 4, 8$).

(a) Construct p (any correct method: Lagrange form or systematic elimination) and simplify to the monomial form.

(b) Evaluate $p(1.5)$ and compare with the true value $2^{1.5} = 2.828427\ldots$

(c) The interpolation error formula gives

$$f(x) - p(x) = \frac{f^{(4)}(\xi)}{4!}\, x(x-1)(x-2)(x-3)$$

for some $\xi \in (0, 3)$. Compute the resulting error bound at $x = 1.5$ (use $f^{(4)}(x) = (\ln 2)^4 2^x$) and verify that the actual error respects it.

Problem 5. Define $S : [0, 2] \to \mathbb{R}$ piecewise by

$$S(x) = \begin{cases} x^3, & 0 \le x \le 1, \\ a + b(x-1) + c(x-1)^2 + d(x-1)^3, & 1 \le x \le 2, \end{cases}$$

on the knots $0, 1, 2$.

(a) Determine the conditions on a, b, c, d for S to be a cubic spline (that is, $S \in C^2[0, 2]$), and solve for a, b, c.

(b) Determine d so that S is a *natural* cubic spline, checking both boundary conditions.

(c) With your values, compute $S(2)$ and $S''(x)$ on each piece, and sketch (or describe) S'': what shape must it have for any natural cubic spline on three knots, and why?